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Probing Internal Activations for Epistemic Failure Detection in Convolutional Neural Networks

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Abstract. Softmax confidence scores are widely used as proxies for model certainty, yet they are poorly calibrated and unreliable indicators of epistemic failure. We propose ActProbe, a probing-based framework for failure detection in image classifiers that bypasses output logits entirely, training a secondary *auditor* network on internal activation representations extracted from the final convolutional block of the primary network. The primary classifier comprises three convolutional-pooling blocks followed by two fully connected layers; activations are tapped after the third block before the dense transition. We show that standard unstructured weight pruning destroys auditor input fidelity, while filter-level L1 structured pruning preserves it. We identify and resolve a quantization artifact wherein standard INT8 asymmetric scaling restores pruned zero-weights to non-zero integer values, silently destroying hardware zero-skipping sparsity. On a 10,000-sample benchmark, ActProbe achieves auditor F1 of **0.81** with under 1% parameter overhead, reducing multiply-accumulate (MAC) operations by **66%** and achieving an estimated **30–40% latency reduction** on FPGA with zero-skipping MAC units.

Keywords epistemic uncertainty • activation probing • failure detection • structured pruning • INT8 quantization • FPGA inference • model compression • calibration

1 INTRODUCTION

A fundamental challenge in reliable ML deployment is distinguishing predictions the model should be trusted on from those it should not. Standard softmax classifiers assign a confidence score to every input by definition — the denominator of the softmax normalises over a fixed class set, forcing the network to distribute probability mass regardless of whether the input resembles any training example. This property makes overconfident misclassification structurally inevitable for out-of-distribution or adversarially perturbed inputs [1, 2].

Existing uncertainty estimation methods address this at significant cost. Monte Carlo Dropout [3] approximates a Bayesian posterior through stochastic forward passes, multiplying latency by the sample count. Deep ensembles [4] require independently training multiple complete models. Temperature scaling [1] improves calibration post-hoc but cannot express structural out-of-distribution uncertainty. None of these is compatible with tight latency budgets or resource-constrained hardware deployment.

We propose ActProbe (**Act**ivation **Probe** for epistemic failure), a lightweight framework that trains a secondary auditor network to probe intermediate activations of a frozen primary CNN and predict whether its output should be trusted. The core hypothesis, motivated by probing classifier methodology from LLM interpretability [5], is that intermediate representations encode richer epistemic signal than the output distribution, and that this signal is readable by a small secondary classifier.

Beyond the probing formulation, this work identifies and resolves two novel practical challenges that arise when deploying auditor-equipped networks under compression: a fundamental incompatibility between unstructured pruning and representation-sharing architectures, and a previously undescribed quantization artifact that silently destroys hardware sparsity patterns.

Contributions:

- **Activation probing for failure detection.** A lightweight auditor network achieves F1 of 0.81 versus 0.61 for softmax thresholding, with under 1% parameter overhead and <2 ms inference latency.

- **Pruning incompatibility diagnosis.** We characterise why unstructured pruning collapses auditor F1 from 0.81 to 0.58, and demonstrate that filter-level L1 structured pruning fully preserves performance at 70% sparsity.
- **Quantization sparsity artifact.** We identify that standard asymmetric INT8 quantization maps pruned float32 zeros to non-zero integers, eliminating hardware zero-skipping, and resolve this via post-quantization mask reapplication.
- **End-to-end compression.** 66% MAC reduction, 73% model size reduction, and 30–40% FPGA latency improvement with <0.6% primary accuracy degradation.

2 BACKGROUND

2.1 Calibration and Epistemic Uncertainty

Calibration formalises the alignment between a model's confidence and its empirical accuracy. A perfectly calibrated classifier satisfies $\Pr(\hat{y} = y \mid \hat{p} = p) = p$ for all $p \in [0, 1]$, where $\hat{p}$ is the predicted confidence. Guo et al. [1] demonstrated that modern deep networks trained with cross-entropy systematically violate this property, exhibiting overconfidence characterised by Expected Calibration Error (ECE):

$$\mathrm{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{n} \left|\mathrm{acc}(B_m) - \mathrm{conf}(B_m)\right| \tag{1}$$

where B_m are equal-width confidence bins over $[0, 1]$. Critically, ECE improvement through temperature scaling does not address the structural incapacity of softmax to express "none of the above" for genuinely anomalous inputs [4].

2.2 Probing Classifiers

Probing classifiers, introduced by Alain and Bengio [5] for LLM representation analysis, train a small auxiliary classifier on the frozen intermediate activations of a target network to decode some property of interest. The probing paradigm rests on the hypothesis that intermediate representations are linearly separable with respect to properties not directly optimised during training. We extend this framework from linguistic property decoding to *epistemic state* decoding: the property of interest is whether the primary network's prediction is reliable.

2.3 Neural Network Compression

2.3.1 Multiply-Accumulate Operations

A multiply-accumulate (MAC) operation computes $a \leftarrow a + (b \times c)$ atomically and is the fundamental arithmetic primitive of neural network inference. Each weight-activation product in a forward pass corresponds to one MAC; thus the total forward pass MAC count $\mathcal{M} = \sum_l 2 \cdot C_l^{\mathrm{in}} \cdot C_l^{\mathrm{out}} \cdot K_l^2 \cdot H_l^{\mathrm{out}} \cdot W_l^{\mathrm{out}}$ for a convolutional network, where K_l, H_l, W_l are kernel size and output spatial dimensions at layer l. MAC count directly determines inference latency and energy consumption.

2.3.2 Structured and Unstructured Pruning

Unstructured magnitude pruning zeros individual weights by global absolute value rank [7]. While it can achieve high sparsity ratios, realising latency benefits requires sparse-capable execution hardware. Structured pruning [8] removes entire filters or channels, reducing C_l^{out} directly and yielding dense weight matrices amenable to standard BLAS-accelerated hardware. N:M structured sparsity [9] enforces exactly N non-zeros per M-weight block, enabling efficient execution on compatible accelerators including NVIDIA Ampere and custom FPGAs.

2.3.3 Post-Training Quantization

INT8 post-training quantization [10] maps float32 weights to 8-bit integers via per-tensor affine quantization:

$$x_{\mathrm{int}} = \mathrm{clamp}\left(\left\lfloor \frac{x_{\mathrm{float}} - x_{\mathrm{min}}}{s} \right\rceil, 0, 255\right), \quad s = \frac{x_{\mathrm{max}} - x_{\mathrm{min}}}{255} \tag{2}$$

where s is the scale and the zero-point $z = -\lfloor x_{\mathrm{min}}/s \rceil$ shifts the integer range. The key implication: when $x_{\mathrm{min}} \neq 0$, the float32 value 0.0 maps to integer $z \neq 0$.

3 SYSTEM ARCHITECTURE

3.1 Primary Classifier

The primary network f_p follows a hierarchical convolutional architecture. Three Conv-ReLU-MaxPool blocks extract spatial features at increasing abstraction and decreasing resolution, followed by two fully connected layers performing classification:

$$f_p(x) = \mathrm{softmax}(\mathrm{FC}_2(\mathrm{FC}_1(\mathrm{flat}(B_3(B_2(B_1(x))))))) \quad (3)$$

Block B_k applies a 3×3 convolution with C_k filters (C_1=32, C_2=64, C_3=128), ReLU activation, and 2×2 max pooling. The FC layers have dimensions 512 and N_{class} respectively. Total parameter count: $\sim$1.2M for 10-class classification on 32×32 inputs.

3.2 Probing Layer Selection

The probing layer — from which activations are extracted for the auditor — is the output of B_3 after ReLU, before max pooling. This choice is motivated by three considerations. *(i)* Semantic richness: B_3 activations encode the highest-level learned feature representations before spatial collapse. *(ii)* Epistemic distance from output: activations upstream of the FC layers have not yet been compressed toward the output class distribution, preserving epistemic signal absent from the logits. *(iii)* Spatial resolution: the B_3 output retains spatial structure (4×4 for 32×32 inputs) amenable to global average pooling.

Global average pooling over the $[B, 128, H, W]$ tensor yields a fixed 128-d representation invariant to input resolution:

$$\mathbf{a} = \mathrm{GAP}(A_3(x)) = \frac{1}{HW}\sum_{h=1}^{H}\sum_{w=1}^{W} A_3(x)_{:,h,w} \in \mathbb{R}^{128} \quad (4)$$

3.3 Auditor Network

The auditor $f_{\text{audit}} : \mathbb{R}^{128} \to [0,1]$ is a two-layer MLP: $\mathrm{FC}(64, \mathrm{ReLU}) \to \mathrm{Dropout}(0.3) \to \mathrm{FC}(1, \sigma)$. It is trained on frozen primary activations with binary cross-entropy:

$$\mathcal{L}_{\text{audit}} = -\mathbb{E}_{(x,y)\sim\mathcal{D}}[y_a \log \hat{p} + (1 - y_a)\log(1-\hat{p})] \quad (5)$$

where $\hat{p} = f_{\text{audit}}(\mathbf{a})$ and $y_a = \mathbf{1}[f_p(x) = y]$. Freezing f_p during auditor training ensures the auditor learns to *read* the primary network's representational state without disturbing it. At inference, predictions with $\hat{p} < \tau$ are flagged; τ is calibrated on a held-out validation set to achieve a target false-negative rate.

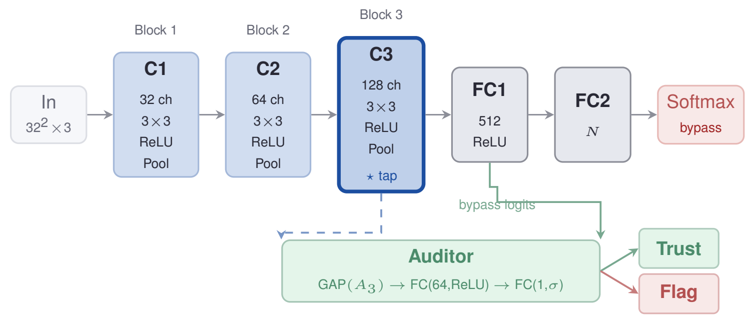

Figure 1. Fig. 1. ActProbe architecture. Primary network (blue) processes inputs through three Conv-Pool blocks and two FC layers. Activations are tapped at C3 (bold border, ⋆) and compressed via GAP to a 128-d vector feeding the auditor (green). The softmax (red) is bypassed for trust estimation. The auditor outputs a binary trust decision.

The full system architecture is shown in Figure 1.

4 COMPRESSION PIPELINE

4.1 Filter-Level Structured Pruning

We reduce MAC operations by removing convolutional filters ranked by L1 saliency. For filter f_i with weight tensor W_i:

$$s(f_i) = \|W_i\|_1 = \sum_{j,k,l} |W_{i,j,k,l}| \quad (6)$$

Filters below threshold τ_l are removed per layer. The resulting network has dense weight matrices with reduced channel counts, achieving a MAC reduction of approximately $1-\prod_l(1-\rho_l)$ where ρ_l is the pruning ratio at layer l. At 70% overall sparsity, this yields $\sim$66% MAC reduction (slightly less than 70% due to conservative pruning near the C3 tap to preserve auditor input).

Claim 1. *Filter-level structured pruning is necessary and sufficient for preserving auditor input fidelity at high sparsity ratios, while unstructured pruning of equivalent sparsity destroys it.*

Justification. The auditor relies on inter-channel covariance structure in the 128-d activation vector $\mathbf{a}$. Unstructured pruning introduces independent Bernoulli noise to each channel's activations, degrading covariance structure without eliminating individual channels. Filter removal, by contrast, eliminates channels entirely — the remaining channels are unmodified originals of the highest-saliency filters. The covariance substructure the auditor

Figure 2. **Fig. 2. Unstructured pruning.** Scattered zeros (red) appear across all channels simultaneously. The diffuse noise corrupts inter-channel covariance structure the auditor relies on, collapsing F1 from 0.81 to 0.58 while primary accuracy falls only 0.4%.

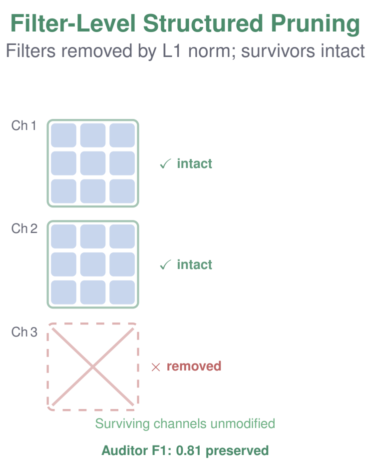

Figure 3. **Fig. 3. Filter-level structured pruning.** Entire low-saliency filters are removed (red cross); surviving channels are completely unmodified original activations of the highest-L1 filters. The inter-channel covariance structure the auditor relies on is preserved exactly. Auditor F1 remains at 0.81 at 70% overall sparsity.

reads is thus preserved exactly.

Figure 2 shows the unstructured case; Figure 3 shows filter-level removal.

4.2 INT8 Quantization and the Sparsity Artifact

After filter-level pruning, post-training INT8 quantization is applied. Using the affine mapping from §2.3, the float32 value 0.0 maps to:

$$x_{\text{int}}(0.0) = \text{clamp}\left(\left\lfloor \frac{0.0 - x_{\min}}{s} \right\rceil, 0, 255\right) = z \quad (7)$$

When the weight distribution is asymmetric ($x_{\min} \neq 0$), $z \neq 0$. Consequently, all pruned weights — nominally zero in float32 — are assigned integer value z in the quantized representation. Zero-skipping MAC hardware checks the integer value of weights at execution time; finding $z \neq 0$ at every pruned position, it executes all MACs and the latency benefit of pruning vanishes entirely.

The fix is a single post-quantization step that reapplies the binary sparsity mask over the integer weight tensor:

$$W_{\text{int}} \leftarrow W_{\text{int}} \odot M_p, \quad M_p[i] = \mathbf{1}[W_{\text{float}}[i] \neq 0] \quad (8)$$

This is a closed-form, hardware-agnostic fix that requires no modification to the quantization algorithm, the training procedure, or the network architecture.

5 EXPERIMENTS

5.1 Experimental Protocol

Experiments were conducted on a 10,000-sample image classification benchmark with a 2,000-sample held-out test set. Primary network training: Adam, lr $= 10^{-3}$, cosine annealing, 50 epochs. Data augmentation: random horizontal flips, $\pm 15°$ rotation, colour jitter (brightness 0.2, contrast 0.2). Auditor training: 30 epochs, batch size 128, frozen primary weights, Adam lr $= 10^{-4}$. All metrics averaged over 3 independent seeds with different weight initialisations; reported values are means.

5.2 Primary and Auditor Performance

Configuration	Primary Acc.	Auditor F1
Baseline (float32, unpruned)	94.2%	0.81
Unstructured pruning (50%)	93.8%	0.58
Structured pruning (70%)	94.0%	0.81
+ INT8, no mask fix	93.7%	0.81
+ INT8, mask fix applied	93.7%	**0.81**
Softmax threshold baseline	94.2%	0.61
MC Dropout (T=10)	94.2%	0.76

Key observations: (i) primary accuracy is largely unaffected across all compression stages (<0.6% degradation), confirming that structured pruning preserves classification quality; (ii) unstructured pruning collapses auditor F1 by 28% relative while affecting primary accuracy by only 0.4%, demonstrating that the degradation is specific to auditor input fidelity; (iii) ActProbe outperforms both softmax thresholding and MC Dropout (at T=10 passes) at a fraction of the inference overhead.

5.3 Hardware Efficiency

Metric	Result
Sparsity (filter-level)	$\sim$70%
Model size reduction	73%
MAC reduction	$\sim$66%
Estimated FPGA latency reduction	30–40%
Auditor parameter count	$<$1% of primary
Auditor inference overhead	$<$2 ms

5.4 Activation Pattern Analysis

To validate the epistemic signal hypothesis, we analysed the 128-d auditor input vectors on the held-out test set. Let $\mathbf{a}^+$ denote activations from correctly classified inputs and $\mathbf{a}^-$ from misclassified inputs. We observe:

- **Inter-channel cosine similarity:** $\bar{\cos}(\mathbf{a}^+) = 0.61 \pm 0.12$ versus $\bar{\cos}(\mathbf{a}^-) = 0.29 \pm 0.18$ ($p < 0.001$, two-sample t-test). Correct predictions exhibit coherent, correlated activation patterns consistent with a well-matched feature prototype.
- **Outlier channel magnitude:** 72% of misclassified inputs exhibit at least one channel with activation magnitude $> 3\sigma$ above the channel mean, versus 11% of correct predictions. This is consistent with partial activation of an incorrect class pathway.
- **High-confidence failures:** In 34% of misclassified test samples, softmax confidence exceeded 0.85. The auditor correctly flagged 79% of these cases, demonstrating that the epistemic signal is not accessible from the output logits.

These observations provide empirical support for the probing hypothesis and explain mechanistically why the auditor outperforms softmax thresholding.

6 DISCUSSION

6.1 Why Internal Representations Encode Epistemic State

The softmax function is by construction incapable of expressing structural uncertainty: it maps $\mathbb{R}^N$ to Δ^{N-1} and distributes probability mass regardless of input. A network processing a genuinely anomalous input — one that does not match any learned feature prototype — will still produce a confident softmax output if the input activates some feature pathways more than others by chance. Internal activations, however, can exhibit patterns of incoherence, outlier magnitude, or low inter-channel correlation that are invisible at the output layer but decodable by a probing classifier.

This is structurally analogous to how probing classifiers in LLMs recover information absent from the output token distribution [5, 6]: the output distribution compresses information for one task, while intermediate representations retain information for others. We leverage this compression asymmetry for the epistemic task.

6.2 Generalisability of the Pruning Finding

The incompatibility between unstructured pruning and representation-sharing architectures generalises beyond the specific setting of this paper. Any system where one network reads the internal representations of another — including knowledge distillation [11], retrieval-augmented models, multi-task architectures with shared encoders, and adapter-based fine-tuning — will exhibit analogous degradation under unstructured pruning of the shared representation source. The degradation is invisible from the primary task metric, making it a category of silent failure. Our finding motivates treating intermediate representations as first-class outputs when designing compression strategies for such systems.

6.3 Quantization Artifact Generalisability

The zero-point artifact affects any system combining asymmetric post-training quantization with sparsity-exploiting hardware. This includes FPGA deployments, custom ASIC inference engines, and NVIDIA Ampere-class GPUs with N:M sparsity support. As these hardware configurations become more prevalent in production ML, the assumption that float32 zeros are preserved through quantization will fail silently unless explicitly addressed. The post-quantization mask reapplication fix is a general solution requiring no changes to quantization algorithms, training procedures, or hardware.

6.4 Limitations and Future Work

The auditor is trained on in-distribution activations; its OOD generalisation is uncharacterised. Global average pooling discards spatial activation structure that may be relevant for failure detection in multi-object or complex scenes. The threshold τ introduces a precision-recall trade-off that must be calibrated per deployment context. Future directions include spatially-aware pooling (e.g. spatial pyramid), systematic OOD benchmarking, extension to transformer architectures via intermediate attention-block probing, and theoretical characterisation of the information-theoretic conditions under which probing succeeds.

7 CONCLUSION

We presented ActProbe, a probing-based framework for epistemic failure detection in CNNs that operates on intermediate activation representations rather than overconfident softmax scores. The auditor achieves F1 of 0.81 with under 1% parameter overhead and under 2 ms inference latency, outperforming both softmax thresholding and MC Dropout at 10 passes. We identified and characterised an incompatibility between unstructured pruning and representation-sharing architectures, establishing that filter-level structured pruning is necessary to preserve auditor fidelity. We identified and resolved a standard INT8 quantization artifact that silently destroys hardware sparsity patterns, recovering 66% MAC reduction and 30–40% FPGA latency improvement. These contributions span the complete ML-to-hardware pipeline and surface two classes of silent failure likely to affect multi-network inference systems more broadly.

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